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21 July 1977
SUPERSEDING
MIL-W-43898A(GL)
30 January 1975

MILITARY SPECIFICATION

SHELTER, MULTI-PURPOSE (MUST)

This specification is approved for use by US Army Natick R&D Command, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers a rigid shelter constructed of aluminum-faced, paper honeycomb core sandwich panels and having the following approximate dimensions: length 12 feet (3.7 m), width 7 feet (2.1 m), height 7 feet (2.1 m). With the use of an insulated fabric enclosure the shelter is expandable to dimensions approximately as follows: length 12 feet (3.7 m), width 10 feet (3.0 m), height 7 feet(2.1 m).

2. APPLICABLE DOCUMENTS

* 2.1 Issues of documents. The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

QQ-P-416	-	Plating, Cadmium (Electrodeposited)
TT-E-527	-	Enamel, Alkyd, Lusterless
TT-E-529	-	Enamel, Alkyd, Semi-Gloss
TT-P-1757	-	Primer Coating, Zinc Chromate, Low Moisture Sensitivity
MMM-A-132	-	Adhesives, Heat Resistant, Airframe Structural, Metal to Metal

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: US Army Natick Research and Development Command, Natick, MA 01760 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 5410

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MIL-C-5541	-	Chemical Conversion Coatings on Aluminum and Aluminum Alloys
MIL-W-6858	-	Welding, Resistance: Aluminum, Magnesium, Non-hardening Steels or Alloys, Nickel Alloys, Heat-Resisting Alloys, and Titanium Alloys: Spot and Seam
MIL-W-8611	-	Welding, Metal Arc and Gas, Steels and Corrosion and Heat Resistant Alloys; Process for
MIL-A-43759	-	Adapters, Airlock, (MUST)
* MIL-A-83377	-	Adhesive Bonding (Structural) for Aerospace Systems, Requirements for
MIL-T-43821	-	Tie-Down Assembly, Cargo (MUST)

STANDARDS

FEDERAL

FED-STD-191	-	Textile Test Methods
FED-STD-595	-	Colors
FED-STD-751	-	Stitches, Seams, and Stitchings

MILITARY

MIL-STD-105	-	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129	-	Marking for Shipment and Storage
MIL-STD-130	-	Identification Marking of U.S. Military Property
MIL-STD-401	-	Sandwich Constructions and Core Materials; General Test Methods
MS 63045	-	Plate, Identification, Miscellaneous

DRAWING

ARMY NATICK RESEARCH AND DEVELOPMENT COMMAND

5-4-1302	-	Shelter, Multi-Purpose (MUST); Drawing List

(Copies of specifications, standards, drawings and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

* 2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

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AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM D1564 - Slab Flexible Urethane Foam

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

(Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

3. REQUIREMENTS

* 3.1 Panel process and sample review. Prior to award of contract, the contractor shall submit to the contracting officer a certified copy of a laboratory test report and a copy of the process specification covering fabrication of the metallic-faced, paper honeycomb core sandwich panels he proposes to use in the construction of the end item. Using his process specification, the contractor shall fabricate a qualification test sandwich panel and shall perform the tests for sandwich panels specified in 4.6.1.1.3. A sample specimen of the test panel, 24 inches (610 mm) by 24 inches (610 mm) minimum, fabricated using 0.040 inch (1.0 mm) aluminum skins, shall be submitted along with the test report and process specification to the contracting officer or his authorized representative for review. The laboratory test report shall contain test data which demonstrates that the panels he proposes to use have been tested and found to comply with all the requirements specified herein. The contractor's process specification shall contain an adequate description for the control of materials, storage conditions prior to use, cleaning of materials, priming or otherwise protecting aluminum skins from corrosion, fabrication, quality standards and control which provide for the surveillance, inspection and testing he proposes to use to manufacture the panels and which will assure continued quality of product. Detailed guidance for the preparation of and requirements to be included in the contractor's process specification is provided in MIL-A-83377.

The adhesive shall be in film form of sufficient thickness to insure good filleting to core at the bond line. The adhesive shall meet all requirements of MMM-A-132, type I, class 2, except metal to metal lap shear strengths shall be as follows:

a. Temperature exposure.

Temperature	Shear strength
Minus 65°F (minus 53.9°C)	3000 psi (20.7 MPa)
Ambient	3000 psi (20.7 MPa)
200°F (93.3°C)	2000 psi (13.8 MPa)

b. Humidity exposure. The adhesive shall withstand exposure to 100 percent relative humidity at 200°F (93.3°C) for two weeks and meet 1500 psi (10.3 MPa) minimum at 200°F (93.3°C).

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c. Salt spray exposure. The minimum strength of the adhesive shall be 2000 psi (13.8 MPa) after exposure to salt spray (20 percent NaCl solution) at 95°F (35°C) for two weeks.

Within thirty days after award of contract, the contractor shall submit to the contracting officer, certified proof that the above tests have been accomplished and satisfactory results obtained.

The purpose of the above requirements is to assist the Government in determining the capability of contractors to manufacture sandwich panels meeting all the requirements of this specification. Notice of completion of review of the specification, laboratory report and test sample under these requirements shall in no way be construed as relieving a contractor from subsequently meeting all the requirements of this specification for the sandwich panels and the end item.

3.2 Alternative components. Components offered as equivalent to components specified hereinafter and on the referenced drawings as a specific manufacturer's part number shall be functionally equal to the manufacturer's part number identified and of equal or better quality. The incorporation and inclusion of such a component in the design of the specified end product shall not require modification or change to any other specified component, and shall not result in increased maintenance to it or any other components, unless such modification or change is specifically approved by the contracting officer. Prior to manufacture of the first article, or if none is required, prior to commencing production, the contractor shall submit for the contracting officer's approval, a list identifying each proposed alternate component, together with proof that each listed component complies with the requirements specified herein. The contracting officer, at his option, may require a physical sample of the proposed substitution for his evaluation. Approval of the submitted listing, together with necessary supporting data, authorizes the commencement of the first article or production manufacture, as applicable, and does not relieve the contractor of the responsibility that these components perform in accordance with specified requirements when incorporated into the end product.

3.3 First article. When specified (see 6.2), the contractor shall furnish a sample of the item to be procured under this specification for first article inspection and approval (see 4.3 and 6.3).

3.4 Initial production testing sample. When specified (see 6.2), the contractor shall furnish samples for initial production inspection and approval (see 4.4 and 6.4).

3.5 Inspection comparison testing sample. When specified (see 6.2), the contractor shall furnish a sample for comparison inspection and approval (see 4.5 and 6.5).

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3.6 Materials and components. Materials and components shall conform to the specifications, standards and drawings listed in section 2 and as specified herein. When materials are not definitely specified, they shall be of the quality normally used by the manufacturer of hard shelters provided the completed item complies with all provisions of this specification.

3.6.1 Sandwich panels. Sandwich panels shall be hot bonded using aluminum face sheets, paper honeycomb core and foam materials specified on Drawing 5-4-1350. Thicknesses of finished panels and core ribbon directions shall be as specified on the drawing. Where core splicings occur the panel shall maintain structural soundness and shall be made either by overlapping or interlocking the honeycomb core segments to be spliced not less than 1 inch (25 mm) nor more than 2 inches (51 mm) or by butting core segments and/or faying surfaces and bonding the surfaces over the entire length of the splice. Where core splicing is accomplished by the overlapping or interlocking of core cells (resulting in crushing of cells) a foaming-type adhesive shall be used to completely fill the entire splice area to prevent any migration of water through the splice. Face sheets, core and edging shall be integrally bonded with adhesive. The fully cured bond shall be capable of meeting the physical requirements specified herein and shall withstand temperatures of minus 65°F to 200°F (-53.8 to 93.3°C) without delamination or separations between face sheets and core.

* 3.6.1.1 Physical properties. Finished panels shall exhibit no surface defects such as bulges, depressions, dents and scratches. Unless otherwise specified on the drawings, tolerances for panel warpage shall be as follows:

Warpage	-	Plus or minus 0.025 inch (0.64 mm) per foot (305 mm) length of panel.

The finished panels shall be resistant to fire, humidity and age and shall conform to the following requirements when tested as specified in 4.6.1.1.3:

Fire resistance	-	Nonburning, or self-extinguishing within 15 seconds after removal of the flame with no smoldering or glowing visible for a maximum of 10 seconds thereafter.
Humidity and age	-	No warping, staining, delamination or other adverse effect on appearance or function.

Face sheet to core bond (4-lb (1.8 kg) Core)

Peel strength, (min) - 4 inch-pounds (0.45 N.m) of torque per inch (25 mm) of width.

Shear beam strength, (min):

TL plane - 175 psi (1.2 MPa)
TW plane - 105 psi (0.72 MPa)

Tension strength - 400 psi (2.76 MPa) or core failure

Compressive strength - 375 psi (2.59 MPa)

Continuity of bond - No void area

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3.6.1.2 Panel identification. Each sandwich panel shall be permanently and legibly marked for identification with a serial number or part number, the lot number, and the date of manufacture. In addition, any panels not fabricated by the shelter manufacturer shall contain a purchase order number and the actual manufacturer's name. Records shall be kept such that each panel can be traced to the lot number of each constituent material.

3.6.2 Fabric enclosure. The fabric enclosure shall be constructed in conformance with the requirements specified on applicable drawings and as specified herein.

3.6.2.1 Polyurethane foam. The polyurethane foam shall be 1-inch (25.4 mm) thick flexible and flame retardant material conforming to the following requirements when tested as specified in 4.6.1.2:

Density, lbs/cu. ft.	2.6 maximum (41.6 kg/m^3)
Indentation load, lbs deflection at 25%	25.0 $\pm$ 5 (89.0 to 133.4 N)
Compression set, 50%	20% maximum
Tear resistance, lbs/lineal inch	2.0 minimum (8.9 N/25.4 mm)
After-flame time, seconds	
Original	2.0 maximum
After oven aging	2.0 maximum
After steam autoclaving	2.0 maximum

3.7 Design and construction. Design and construction of the shelter shall conform to the drawings listed on Drawing 5-4-1302 and as specified hereinafter. The unit shall be free of panel delaminations and shall meet all physical and environmental requirements specified herein when tested as specified in 4.6.3.3.

3.7.1 Leveling, opening and setting up. There shall be no difficulty in leveling, opening and setting up the unit and attaching the fabric enclosure (see 4.7.5).

3.7.2 Internal air leakage. Internal air leakage from the unit shall not exceed 30 standard cubic feet per minute (scfm) (0.01 m^3/s) in the closed condition, and 60 scfm (0.03 m^3/s) in the open configuration when pressurized to 0.3 inch of water (water gauge) (74.7 Pa) when tested as specified in 4.7.6.

3.7.3 Blackout. There shall be no internal light visible from outside of the unit in either the open or closed configuration when tested as specified in 4.7.7.

3.7.4 Structural soundness. The unit shall maintain structural soundness and shall show no evidence of delamination, deformation or damage, loosening or weakening of structural parts when subjected to the following:

a. When fully loaded (6500 pound (2.9 Mg) dummy load):

(1) Towed on its skids (see 4.7.8).
(2) Transported on a demountable dolly set (see 4.7.9 and 6.7).
(3) Subjected to railroad humping (see 4.7.10 and 6.7).

b. When loaded with a dummy load to a gross weight of 25,000 pounds (11.3 Mg) (container and dummy load) and hoisted by the rings (see 4.7.11).

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3.7.5 Environmental characteristics.

3.7.5.1 Water leakage. The shelter shall show no accumulation of water when tested under a simulated rainfall of 4 ± 1 inches (101.6 ± 25 mm) per hour as specified in 4.7.12.

3.7.5.2 Low temperature. There shall be no delamination of panel surfaces or separations of edge extrusions or permanent warping when the unit is tested at a temperature of minus 65° ± 5°F (-53.9 ± 3°C) for 24 hours, with the internal container temperature maintained at 70° ± 5°F (21.1 ± 3°C) as specified in 4.7.13.

3.7.5.3 Humidity. There shall be no delamination of panel surfaces or accumulation of moisture within the shelter when the unit, in its closed configuration, is tested at a temperature of 100° ± 5°F (37.8 ± 3°C) with relative humidity of 97 ± 2 percent as specified in 4.7.14.

3.7.5.4 Atmospheric pressure. The unit, or individual sub-assembled panels equivalent to a complete unit, shall show no evidence of damage to the panels or structure when tested at atmospheric pressures encountered up to 50,000 feet (15.2 km) above sea level (87.5 mm of mercury (11.7 kPa) as specified in 4.7.15.

3.7.5.5 Solar load. The unit shall show no evidence of delamination, buckling or other deterioration or permanent deformation when tested at simulated solar loads sufficient to raise the outer skin temperatures of roof sections (fixed and folding) to 200° ± 5°F (93.3 ± 3°C) as specified in 4.7.16.

3.7.6 Installation of closeouts and inserts. Installation of edge extrusions, closeouts, and inserts shall be as shown on the drawings and as specified herein. Aluminum alloy edge extrusions shall be cleaned and prepared for bonding in accordance with the cleaning, preparation and handling methods specified for the aluminum alloy face sheets used in the manufacture of the sandwich panels. Phenolic edging shall be cleaned on the side to be bonded by a light sanding to remove foreign matter and to roughen the surface for bonding. Steel inserts shall be clean and free of dirt, oil, grease or other foreign matter. Prior to assembly of edge members, closeouts and inserts to the sandwich panels, sufficient adhesive shall be applied to insure complete protection of honeycomb paper core from moisture penetration and to insure total contact with edge members and inserts to be applied. Both sides of the aluminum edge extrusion leg and the faying surfaces of the extrusions shall be coated with the adhesive specified on the drawings. Use of the adhesive shall be in accordance with the adhesive manufacturer's instructions for application, curing, pot life and age. Excess adhesive shall be removed from joints. No gaps shall be visible between aluminum faces and extruded or phenolic edges or closeouts and at the intersection of the extruded edges or closeouts.

3.7.6.1 Inserts. All inserts shall have the dimensions, shape, and be located as specified on Drawing 5-4-1318. When tested as specified in 4.7.17 and 4.7.18, the inserts shall withstand the torque and pull-out loads specified in table I without failure of the insert, panel, or potting compound.

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TABLE I. Insert strength

Insert (thd size-dia)	Pull-out		Torque	
	(lb)	(kN)	(ft-lbs)	(N· m)
1/4-28, 11/16	1500	6.7	25	33.9
5/16-24, 11/16	1500	6.7	25	33.9
3/8-24, 3/4	1500	6.7	50	67.8
1/4-28, 7/16	1000	4.4	20	27.1
5/16-24, 1/2	1000	4.4	20	27.1
3/8-16, 9/16	1500	6.7	40	54.2

3.7.7 Stitching of cloth components. Stitching of fabric enclosure and blackout curtains shall conform to type 301 of FED-STD-751, 5 to 7 stitches per inch (2 to 3 stitches per cm) where indicated on the drawings. The ends of stitching shall be overstitched not less than 1 inch (2.5 cm) except where the ends are turned under in a hem, held down by other stitching or cementing. Thread breaks shall be overstitched not less than i inch (2.5 cm) at each break. Thread breaks noted during inspection shall be repaired by overstitching the existing stitching for a distance of 1 inch (2.5 cm) on both sides at each break. The ends of repair stitchings are not required to be backstitched. All thread ends shall be trimmed to 1/2 inch (1.3 cm) or less. The addition of any lubrication to the polyester thread prior to or during the sewing operation is prohibited.

3.7.8 Cementing of cloth components. Cementing of seams, patches and reinforcements shall be as specified on the drawings. Use of the adhesive in cementing shall be in accordance with the adhesive manufacturer's instructions for cleaning, application (including humidity and temperature conditions), curing, pot life and age. Cemented areas shall be free of puckers, loose edges, unadhered pockets or tunnels. There shall be no unsightly smearing or spreading of adhesive, and surrounding areas shall be kept clean and free from excess adhesive. Stains and discolorations of the coated cloths shall not show beyond 1/4 inch (0.6 cm) of the cemented area.

3.8 Welding. All welding shall be as specified on the drawings. Materials to be welded shall be clean and free from scale, corrosion, oil, water and other matter. Welds shall be sound, smooth, free from flash and shall show good fusion.

3.8.1 Qualification of welders. All welding operations shall be performed by welders or welding operators who have met the requirements referenced in MIL-W-8611 and MIL-W-6858.

3.9 Finish of metal components.

3.9.1 Surface preparation.

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3.9.1.1 Steel. All steel components and subassemblies to be painted shall first be clean and free of grease, oil, dirt and other foreign matter and shall then be cadmium plated in accordance with type III, class 2 of QQ-P-416.

3.9.1.2 Aluminum. Prior to painting, all aluminum surfaces shall be cleaned and prepared for painting in accordance with class 3 of MIL-C-5541.

3.9.2 Priming. After surface preparation, all exposed surfaces, except the interior floor and that portion of the interior wall of the folding side which becomes a floor, shall be coated with primer conforming to composition G or L of TT-P-1757, color Y or T. The interior floor area and that portion of the folding side panel which becomes a floor shall be primed as specified on Drawing 5-4-1308.

3.9.3 Enameling. When primer is dry, all exterior surfaces shall be coated with enamel conforming to TT-E-527, olive drab color No. 34087 of FED-STD-595 and all interior surfaces, except the floor area and side portion of the folding panel which becomes a floor, shall be coated with enamel conforming to TT-E-529, pale green color No. 24534 of FED-STD-595. The interior floor area and that portion of the side panel which becomes a floor shall be finished as specified on Drawing 5-4-1308. The coating shall level out to an adherent, continuous and uniform film without runs, wrinkles, streaks or areas of no film. Any coating damaged during assembly or examination shall be touched up.

3.10 Marking.

3.10.1 Identification. Shelters shall be marked for identification in accordance with MIL-STD-130. Location of the markings shall be as shown on the applicable drawings.

3.10.2 Nameplate. A nameplate, in the size specified and containing the data specified in MS-63045, shall be attached to the unit in the location indicated on applicable drawing.

3.11 Government furnished components.

3.11.1 Adapter, air lock. Each shelter shall be provided with an air lock adapter conforming to type I of MIL-A-43759. When specified (see 6.2), air lock adapters shall be furnished by the Government.

3.11.2 Tie down assembly, cargo. Each shelter shall be provided with a cargo tie-down assembly conforming to MIL-T-43821. When specified (see 6.2), the tie-down assembly shall be furnished by the Government.

3.11.3 Transporting equipment.

3.11.3.1 Dolly set, lift type. The transportable shelter lift type dolly set shall be furnished by the Government (see 3.7.4 and 6.2).

3.11.3.2 Truck. The truck required for use in the transporting test shall be a standard 2-1/2 or 5-ton cargo truck and shall be furnished by the Government (see 3.7.4 and 6.2).

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* 3.12 Durability. The complete multi-purpose shelter shall be capable of withstanding a durability test of 1440 hours without failure when tested in accordance with 4.7.19.

3.13 Workmanship. All metal forming shall be done with dies or fixtures to insure uniformity of size and shape. Sheet metal or structural sections used in fabrication of the shelters shall not be kinked, bent, or have ripples. Shearing and clipping shall be done accurately and cleanly on all portions of the work and shall be neatly finished. All bolt and rivet holes shall be accurately and cleanly punched or drilled. Rivets, when used, shall be expanded to completely fill the holes, with full-sized heads neatly finished and in full contact with the surface of the member. Rivets shall be dipped in sealant prior to expanding, as noted on applicable drawings. Bolts, nuts, and screws shall be drawn tight and so secured that they shall not work loose under vibration. All surfaces shall be free of burrs, slivers, rough or sharp edges. Parts shall not be fractured, split, punctured, dented or malformed. Parts shall be in visual alignment and securely assembled. All functioning components shall operate as intended without abnormal force. Welds shall not be cracked and shall be free from slag inclusion, undercut, scale, or flux deposits. Accessories shall be securely fastened and braced to prevent damage or loosening during units' intended use or transportation. The sheet metal in the finished unit shall be as flat as stacked sheet metal. There shall be no areas of rust. Finish shall be free of blistering, peeling, chips, and areas of no-film. Gaskets shall not be torn, cut or split and shall be free of finish. Cloth components shall be clean and free of holes, cuts or tears. Seam allowances shall be maintained with seam properly sewed so that no open seams, twists or pleats result.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of inspection. The inspection requirements specified herein are classified as follows:

(a) First article inspection (see 4.3).
(b) Initial production sample inspection (see 4.4).
(c) Comparison sample inspection (see 4.5).
(d) Quality conformance inspection (see 4.6).

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4.3 First article inspection. When a first article is required (see 3.3 and 6.2), the complete shelter shall be examined for defects listed in table III, dimensions specified, and tested as specified in 4.7.4.5, 4.7.5, 4.7.6, table IV, 4.7.17.2 and 4.7.18.2. The presence of any visual defect, any dimension not within the specified requirements or failure to comply with any test requirement shall be cause for rejection of the initial production sample.

* 4.4 Initial production sample inspection. When an initial production sample is required (see 3.4 and 6.2), the complete shelter shall be examined for defects listed in table III, dimensions specified, and tested as specified in 4.7.4.5, 4.7.5, 4.7.6, table IV, 4.7.17.2 and 4.7.18.2. One complete multi-purpose shelter shall be tested as specified in 4.7.19. The presence of any visual defect, any dimension not within the specified requirements or failure to comply with any test requirement shall be cause for rejection of the initial production sample.

* 4.5 Comparison sample inspection. When a comparison sample is required, the complete shelter shall be examined for defects listed in table III, dimensions specified, and tested as specified in 4.7.4.5, 4.7.5, 4.7.6, table IV, 4.7.17.2 and 4.7.18.2. One complete multi-purpose shelter shall be tested as specified in 4.7.19. The presence of any visual defect, any dimension not within the specified requirements or failure to comply with any test requirement shall be cause for rejection of the comparison sample.

4.6 Quality conformance inspection. Except as otherwise specified herein, sampling for inspection shall be in accordance with MIL-STD-105.

4.6.1 Component and material inspection. In accordance with 4.1 above, the components and materials shall be inspected in accordance with all requirements of referenced specifications, drawings and standards unless otherwise excluded, amended, modified or qualified in this specification or applicable purchase document.

* 4.6.1.1 Sandwich panels. Inspection shall be made of the sandwich panels specified in 3.6.1. Inspection records of all examinations and tests shall be maintained for each lot of panels and a certified copy of the inspection report for each production lot of sandwich panels shall be submitted to the contracting officer or his authorized representative. Each report shall verify that the inspections specified have been performed and complied with, shall include the actual test values obtained and the identity of the production lot represented by the tests performed. The inspection lot shall consist of one day's production not to exceed a maximum of 30 panels and the sample unit shall be one complete sandwich panel.

4.6.1.1.1 Visual examination. Sandwich panels shall be examined for visual defects. Any bulge, depression, dent or scratch shall be classified as a defect. The inspection level shall be II and the acceptable quality level (AQL), expressed in terms of defects per hundred units, shall be 6.5.

4.6.1.1.2 Dimensional examination. Sandwich panels shall be examined for defects in dimensions. Any dimension not within the specified requirements shall be classified as a defect. The inspection level shall be S-3 with an AQL of 4.0, expressed in terms of defects per hundred units.

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4.6.1.1.3 Testing of sandwich panels. One panel, or portion of panel, from each production lot shall be tested as specified in table II for the characteristics shown. Results of each test shall be reported as pass or fail and failure of the sample panel to meet any of the specified requirements shall be cause for rejection of the production lot.

4.6.1.1.4 Sandwich panel, with inserts. One complete sandwich panel shall be randomly selected from the initial lot of panels. The inspection lot shall be of the size specified in 4.6.1.1. Four of each different size and diameter inserts indicated in table I (total of 24 inserts) shall be potted in this panel using the same techniques and materials used in fabricating the shelter. These inserts shall be located a minimum of six inches (152 mm) in from panel edges and six inches (152 mm), on center, from each other. Two identical inserts shall be tested as specified in 4.7.17.1 with the other two identical inserts tested as specified in 4.7.18.1. If failure of one or more inserts occurs during either one or both tests, a new panel, chosen randomly from the same inspection lot, shall have 12 or 24 inserts potted in, as applicable, and tested as specified above. This procedure shall be repeated until no failures occur. All failures occurring during this testing shall be recorded, including the size and diameter of insert(s) failing and the failure mode.

TABLE II. Sandwich panel testing

Characteristic	Requirement paragraph	Test method or paragraph	Number of determinations per sample panel
Fire resistance 1/	3.6.1.1	4.7.1	1
Humidity 1/	3.6.1.1	4.7.2	1
Age 1/	3.6.1.1	4.7.3	1
Face sheet to core bond:			
Peel strength	3.6.1.1	4.7.4.1	2
Shear strength:	3.6.1.1	4.7.4.2	
TL direction			1
TW direction			1
Tension strength	3.6.1.1	4.7.4.3	2
Compressive strength	3.6.1.1	4.7.4.4	2
Continuity of bond	3.6.1.1	4.7.4.5	-

1/ Testing of panels for fire resistance, humidity and age shall be performed on the initial lot of panels only.

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4.6.1.2 Polyurethane foam. Polyurethane foam shall be tested for compliance with requirements specified in 3.6.2.1. The lot shall be all the foam offered for inspection at one time. Sampling and testing shall be as specified in the applicable test method. Any nonconformance shall cause rejection of the lot.

Characteristic	Test method
Density	ASTM D1564 Sec 69-74
Indentation load, deflection	Sec 19-25
Compression set, 50%	Sec 12-18
Tear resistance, lbs/linear inch	Sec 75-81
After flame time	
Original	FED-STD-191 No. 5905
After oven-aging (after 22 hrs @ 284°F (140°C))	ASTM D1564 Sec 45-49.2
After steam autoclaving (5 hrs @ 250°F (121°C))	Sec 5-9.5

4.6.2 Process examination. Examination shall be made of the following fabrication processes to establish conformance with specified requirements. Whenever nonconformance is noted, correction shall be made to the process and all items processed.

a. That aluminum edge extrusions, phenolic closeouts, and inserts are cleaned and prepared prior to bonding for compliance with 3.7.6.

b. That sufficient adhesive is applied to sandwich panel or extrusion to insure compliance with 3.7.6.

c. That both sides of aluminum edge extrusion legs, the faying surfaces of the extrusions and panel skins are coated with adhesive for compliance with 3.7.6.

d. That use of the adhesive for bonding of edge extrusions, closeouts and inserts is in accordance with the adhesive manufacturer's instructions for compliance with 3.7.6.

e. That insert size, locations, and shape are as specified in 3.7.6.1.

f. That no lubrication is applied to the polyester thread prior to or during sewing for compliance with 3.7.7.

g. That use of the adhesive for the fabric enclosure is in accordance with the adhesive manufacturer's instructions for compliance with 3.7.8.

h. That materials to be welded are free from scale corrosion, oil, water and other foreign matter (see 3.8).

i. That materials to be cadmium plated are free from grease, oil, dirt and other foreign matter (3.9.1.1).

j. That steel components are cadmium plated, prior to painting, for compliance with 3.9.1.1.

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k. That aluminum surfaces are chemically treated, prior to priming, for compliance with 3.9.1.2.

l. That all exposed surfaces to be painted are prepared and primed prior to enameling (see 3.9.2).

m. That forming of metal parts, shearing and clipping of sheet metal, bolt and rivet holes are in compliance with 3.13.

* 4.6.3 End item inspection. The sample unit shall be one completely fabricated shelter. The inspection lot shall consist of all shelters offered for inspection at one time, except as otherwise specified in 4.6.3.3.

4.6.3.1 Visual examination. Shelters shall be examined for defects in table III. The inspection level shall be II with an AQL of 6.5 for major defects and 10.0 for total major and minor defects combined, expressed in terms of defects per hundred units.

TABLE III. Classification of defects for visual characteristics

Examine	Defect	Classification Major	Minor
Finish	Not as specified; not finished where required	X	
	Not adherent, e.g., blistered, peeled	X	
	Runs, wrinkles, streaks or areas of no film		X
	Finish not dry, i.e., wet or tacky to touch		X
	Scratch, abrasion exposing prime coat or bare metal		X
	Any area of rust	X	
Design, construction and workmanship	Any design characteristic not in accordance with specified requirement	X	
	Component or part fractured, split, punctured, dented or malformed	X	
	Component missing, inoperative or will not operate as intended	X	
	Component not properly assembled or secured	X	
	Any operation omitted or not properly performed	X	
	Any functioning component which requires abnormal force to operate	X	
	Burr, rough or sharp edge, sliver which may be injurious to personnel or cargo	X	

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TABLE III. Classification of defects for visual characteristics (cont'd)

Examine	Defect	Classification Major	Classification Minor
Design, construction and workmanship (cont'd)	Gaskets torn or split	X	
	Gaskets or seals painted or otherwise coated where serviceability may be affected		X
	Visible gaps between aluminum faces or extruded edges or closeouts	X	
Welding and brazing (where required)	Not type specified, missing where required	X	
	Incomplete, burn through area, pits, crack or fracture, or otherwise not fused	X	
	Slag inclusion, undercut, not smooth and uniform, scale or flux deposit not removed	X	
Metal fasteners, bolts, nuts, screws, studs	Missing, not type specified, broken, stripped or loose	X	
	Lockwasher missing where required		X
Rivets	Missing	X	
	Not type specified	X	
	Not sufficiently peened, not drawn tight, excessively peened		X
Cloth	Cut or tear longer than 1/8 inch (0.3 cm)	X	
	Hole larger than 1/8 inch (0.3 cm) diameter	X	
Seams and stitching	Open seam:		
	- 1 inch (2.5 cm) or less		X
	- more than 1 inch (2.5 cm)	X	
	NOTE: A seam shall be classified as open when one or more stitches joining a seam are broken, or when one or more skipped or runoff stitches occur. Thread breaks not overstitched shall be classified as open seams.		
Stitching	Overstitched less than 1 inch (2.5 cm) (except where ends are held down by other stitching or cementing or turned under in a hem)		X

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TABLE III. Classification of defects for visual characteristics (cont'd)

Examine	Defect	Classification Major	Classification Minor
Seam type	Wrong seam type	X	
Stitch type	Wrong stitch type	X	
Hems	Twisted or pleated		X
Thread ends	Not trimmed to 1/2 inch (1.3 cm) or less		X
Cementing of cloth components	Unadhered pocket, tunnel		X
	Loose edge or unadhered seam	X	
	Stain or discoloration beyond 1/4 inch (0.6 cm) of cemented area		X
Slide fastener	Slider jams or fails to interlock chain	X	
Marking	Missing, incomplete, not legible, not specified type or size, misplaced		X

4.6.3.2 Dimensional examination. Examination of the shelter shall be made for defects in dimensions. Any dimension not within the specified requirements shall be classified as a defect. The inspection level shall be S-2 with an AQL of 4.0, expressed in terms of defects per hundred units.

4.6.3.3 Testing of the end item. Each shelter shall be tested for panel delamination, operation, and internal air leakage as specified in 4.7.4.5, 4.7.5 and 4.7.6, respectively. Any nonconformance shall be cause for rejection of the unit. One shelter selected at random from each lot, and already tested as specified in 4.7.4.5, 4.7.5 and 4.7.6 shall be tested as specified in table IV except that transporting and railroad humping tests shall not be performed. A lot for this inspection shall consist of 25 shelters or less. Any nonconformance shall be cause for rejection of the lot.

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TABLE IV. Testing of the end item

Characteristic	Requirement paragraph	Test method or paragraph
Blackout	3.7.3	4.7.7
Structural soundness:	3.7.4	
Towing		4.7.8
Transporting:		
Dolly set		4.7.9
Railroad humping		4.7.10
Hoisting		4.7.11
Water leakage	3.7.5.1	4.7.12
Low temperature	3.7.5.2	4.7.13
Humidity	3.7.5.3	4.7.14
Low pressure	3.7.5.4	4.7.15
Solar load	3.7.5.5	4.7.16

Examination for panel delamination shall be conducted upon completion of each characteristic test and shall be performed as specified in 4.7.4.5. Since the presence of delamination is cause for rejection of the unit and the lot, any further testing required shall be discontinued.

Testing for transporting and railroad humping requirements shall be conducted only on the preproduction sample unit or initial unit of production.

* 4.6.4 Packaging inspection. An examination shall be made of each end item to determine that packaging and marking are in compliance with the requirements of section 5. Any component missing, any opening not covered and secured as specified or any markings not as specified shall be cause for rejection of the unit.

4.7 Test methods.

4.7.1 Fire resistance, sandwich panels.

4.7.1.1 Test specimen. Test specimens of metallic-faced, paper honeycomb core sandwich panels without edging shall be approximately 4 inches by 8 inches (102 by 203 mm).

4.7.1.2 Apparatus. Apparatus shall consist of a standard 9.5 mm barrel Bunsen or Tirrill burner fitted with a 48 ± 1 mm wide wing top and any suitable holding devices which will support the specimen horizontally and allow an unobstructed expanse of a minimum of 6 inches (152 mm) of the specimen (a typical arrangement may consist of two laboratory ring stands with adjustable flat-surface clamps which may be locked in place by set screw lock nut).

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4.7.1.3 Procedure. Place the specimen on its supports and at an appropriate level for the test. Adjust the burner with the wing top to provide a blue flame whose visible portion is 1-1/2 inches (38 mm) high with clearly defined inner cone 1/4 inch (6 mm) high. Place the burner under the specimen so that the flame impinges transversely across the 4 inch (102 mm) width of the specimen surface. After one minute, remove the flame. The specimen shall be judged "nonburning" if there is no evidence of ignition. If ignition should occur, note the time, in seconds, when the flame extinguishes and when smoldering or glowing is no longer visible (see 3.6.1.1).

4.7.2 Humidity, sandwich panels.

4.7.2.1 Test specimen. The test specimen shall be, approximately, a 12 inch by 12 inch (305 by 305 mm) piece of metallic-faced, paper honeycomb core sandwich panel without edging.

4.7.2.2 Apparatus. Apparatus shall consist of a suitable test chamber capable of maintaining the relative humidity and temperature required for the test.

4.7.2.3 Procedure. The test specimen shall be placed in the test chamber and subjected to an environment of 97 ± 2 percent relative humidity at a temperature of 120° ± 5°F (48.9 ± 3°C) for a period of seven days. At the end of the period the specimen shall be examined for compliance with 3.6.1.1.

4.7.3 Accelerated aging, sandwich panels.

4.7.3.1 Test specimen. The test specimen shall be, approximately, a 12 inch by 12 inch (305 by 305 mm) piece of metallic-faced, honeycomb paper core sandwich panel without edging.

4.7.3.2 Apparatus. Apparatus shall consist of a suitable oven capable of maintaining the temperature required for the test.

4.7.3.3 Procedure. The test specimen shall be placed in the oven and subjected to a temperature of 200° ± 5°F (93.3 ± 3°C) for a period of seven days. At the end of the period the specimen shall be examined for compliance with 3.6.1.1.

4.7.4 Face to core bond, sandwich panel.

4.7.4.1 Peel strength.

4.7.4.1.1 Test specimen. Each test specimen shall be 3 ± 1/16 inches (76.2 ± 1.6 mm) wide and approximately 12 inches (305 mm) long. Core ribbon direction shall be lengthwise of the specimen.

4.7.4.1.2 Apparatus. The apparatus for the test shall be of the climbing drum peel type as described for the sandwich peel test specified in MIL-STD-401.

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4.7.4.1.3 Procedure and computation. The test procedure shall be as specified for sandwich peel test specified in MIL-STD-401. Computation of test results, to determine compliance with 3.6.1.1 shall be as specified in MIL-STD-401 except that the torque necessary to bend metal facing shall be determined and subtracted from the average torque needed to peel the sandwich.

4.7.4.2 Shear beam strength.

4.7.4.2.1 Test specimen. Each test specimen shall be 3 ± 1/16 inches (76.2 ± 1.6 mm) by approximately 8 inches (203 mm) long with the core ribbon direction (L) lengthwise to the length of the panel, or transverse (W), as required.

4.7.4.2.2 Apparatus. Apparatus shall consist of three loading blocks; one loading block 3 inches by 2 inches by 3/16 inch (76.2 by 50.8 by 1.6 mm) thick steel sheet and two reaction blocks 3 inches by 2 inch by 3/16 inch (76.2 by 50.8 by 1.6 mm) thick steel sheet; three self-aligning "V" blocks, and a testing machine capable of applying a load at a constant rate of movement.

4.7.4.2.3 Procedure. The test specimen shall be set up for center-point loading on a 6 inch (152 mm) span. The specimen shall be loaded in bending at a constant rate of movement of the movable head of the testing machine of not less than 0.025 (0.01) and not more than 0.050 inch per minute (0.02 mm/s) until failure occurs. Testing will be performed at room temperature.

4.7.4.2.4 Computation. The face to core bond shear strengths shall be calculated to determine compliance with 3.6.1.1 using the following formula:

$$F_s = \frac{L}{2(\text{Specimen width})\ (\text{Thickness of sandwich panel})}$$

where: F_s = shear stress (psi)

L = applied load

4.7.4.3 Tension strength, sandwich panel. Sandwich panel tension tests shall be performed as specified in MIL-STD-401 to determine compliance with 3.6.1.1 with specimen sizes of 2 inches by 2 inches (51 by 51 mm).

4.7.4.4 Compression strength, sandwich panel.

4.7.4.4.1 Test specimen. Each test specimen shall be 4 inches by 4 inches (102 by 102 mm).

4.7.4.4.2 Apparatus. Apparatus shall consist of a testing machine that is capable of applying a load at a constant rate of movement.

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4.7.4.4.3 Procedure. The specimen shall be loaded uniformly in flatwise compression over the entire surface of the specimen at a constant rate of .003 inch per minute (0.001 mm/s) ± 25% until failure occurs. The load shall be applied to the specimen through blocks of the self-aligning type and the test shall be performed at room temperature with dry specimens.

4.7.4.4.4 Computation. Flatwise compressive strength valves (used to determine compliance with 3.6.1.1) shall be calculated as follows:

$$C = P/A$$

where: C = Compressive strength in psi (Pa)
P = Ultimate load in pounds (N)
A = Area of test specimen in square inches (m^2)

4.7.4.5 Continuity of bond, sandwich panel. Continuity of face to core bond shall be determined by tapping the face sheets over the bond areas with a coin or small plastic hammer to produce the sound effect for detecting a void area. A comparatively dull sound is indicative of a void or delaminated area.

4.7.5 Operational test. The unit shall be placed on fairly even ground and leveled through the use of the jacks. The sides shall be opened and the fabric enclosure shall be attached. A complete cycle of leveling, opening and closing of the shelter shall be conducted to determine compliance with the operational requirements of 3.7.1.

4.7.6 Internal air leakage.

4.7.6.1 Closed configuration. With all the openings closed and the anti-backdraft damper taped or pegged closed, an air supply shall be furnished to the unit and an internal pressure of 0.3 inches of water (water gauge) (74.7 Pa) shall be obtained and maintained. The pressure and flow shall be stabilized. The amount of air (scfm) (m^3/s) supplied and required to maintain the specified internal pressure shall be recorded to determine compliance with 3.7.2.

4.7.6.2 Open configuration. With all the openings closed, the anti-backdraft damper taped or pegged closed and the fabric enclosure in place, the unit shall be subjected to an internal air pressure of 0.3 inches of water (water gauge) (74.7 Pa) shall be obtained and maintained. The amount of air (scfm) (m^3/s) supplied and required to maintain the specified internal pressure shall be recorded to determine compliance with 3.7.2.

* 4.7.7 Blackout. The expanded unit shall be placed in a dark environment. A 40 watt fluorescent light shall be suspended from the ceiling brackets of the unit. With all covers in place and doors closed, and blackout curtains over all windows, observe the unit from a distance of 30 feet (9.1 m) to determine compliance with 3.7.3.

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4.7.8 Towing. The unit, in its closed configuration and with an internal dummy load of 6500 pounds (2.9 Mg) uniformly distributed over the floor area, shall be towed over rough terrain on its skids for a distance of 1000 feet (305 m) forward and 1000 feet (305 m) backward at a speed of 5 ± 2 miles per hour (8.0 ± 3.2 km/hr). At the end of the test the unit shall be examined for compliance with 3.7.4.

4.7.9 Transporting. The closed unit, with internal dummy load of 6500 pounds (2.9 Mg), uniformly distributed over the floor area, shall be transported over road surfaces in accordance with the roadability requirements listed below. The unit shall be towed on a transporter dolly (see 6.7).

Type of surface and speed	Distance each test
Belgian block - 15 to 20 mph (24.1 to 32.2 km/hr)	50 miles (80.5 km)
Graded gravel road - 5 sudden stops from 15 mph (24.1 km/hr)	
Two inch washboard - 5 mph (8.0 km/hr)	4100 feet (1.2 km)
Two to four inch radial washboard - 15 mph (24.1 km/hr)	1125 feet (0.3 km)
Three inch spaced bumps - 20 mph (32.2 km/hr)	1300 feet (0.4 km)
Cross country No. 1 (moderate) (Speed shall vary from 5 to 20 mph (8.0 to 32.2 km/hr)	50 miles (80.5 km)

The unit shall be examined for compliance with 3.7.4.

4.7.10 Railroad humping. The shelter unit with its internal dummy load securely restrained shall be loaded on a standard flatcar in accordance with normal railroad loading practice (see fig. 1). The test shall be conducted on a flat stretch of track with the test car traveling a minimum of 8.5 miles per hour (13.7 km/hr) upon impact against one boxcar and one gondola car with air brakes set, and the gondola loaded to 20 tons (18 Mg). The test shall be repeated with the impact being made in the opposite direction. The unit shall be examined for compliance with 3.7.4.

4.7.11 Hoisting. The shelter unit shall be tested with an internal dummy load evenly distributed over the floor area, sufficient to produce a gross container weight of 25,000 pounds (11.3 Mg). The loaded unit shall then be hoisted off the ground by means of a four-cable sling, having cable lengths such as to form approximate 45 degree angles to the roof plane when attached to the lifting rings of the unit and the source used to lift the unit. The unit shall be held 6 to 10 inches (152 to 254 mm) above the ground for a minimum of 30 minutes. At the end of the period, the unit, floor and rings shall be examined for compliance with 3.7.4.

4.7.12 Water leakage. The shelter and its fabric enclosure shall be exposed to a simulated rainfall of 4 ± 1 inches (101.6 ± 25 mm) per hour as measured by a U.S. Weather Bureau type gauge. The direction of the rainfall shall be angled at 45 degrees from the vertical. The roof and each of the sides shall be exposed to this angled rainfall for 30 minutes. The rainfall shall be dispersed uniformly

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over the entire test area. This test shall be performed with the unit in the closed and in the expanded configurations. All openings, doors and panels shall be closed and latched. At the completion of the tests, the inside of the unit shall be examined for evidence of accumulation of water to determine compliance with 3.7.5.1.

4.7.13 Low temperature. The expanded shelter with the fabric enclosure in place shall be placed in a cold chamber. The chamber temperature shall be lowered to minus 65° ± 5°F (-53.9 ± 3°C), and the shelter allowed to remain at this temperature for 24 hours. Heat shall be supplied to the shelter at a rate that will raise the internal temperature of the shelter to 70°F (21.1°C) in two hours. Temperature measuring instruments shall be located midway between the floor and ceiling and one foot from a wall. The temperature inside the shelter shall be maintained at 70° ± 5°F (21.1 ± 3°C) for a period of two hours. The chamber shall be allowed to return to room temperature after which the shelter shall be examined for compliance with 3.7.5.2.

4.7.14 Humidity. The shelter shall be placed in the test chamber in the closed configuration, with all covers in place and doors latched. The temperature and relative humidity shall be gradually raised to 100° ± 5°F (37.8 ± 3°C) and 97 ± 2 percent respectively and allowed to remain under these conditions for 24 hours. The unit shall then be allowed to return to prevailing room conditions either within or upon removal from the test chamber and within one hour shall be examined to determine compliance with 3.7.5.3.

4.7.15 Low pressure (altitude). The closed shelter with all openings closed (except that the anti-backdraft damper shall be free to operate) shall be placed in an altitude chamber at ambient temperature. Reduce the chamber internal pressure to 87.5 mm of mercury (11.7 kPa). (Corresponding to an altitude of 50,000 feet (15.2 km.) Maintain the chamber pressure for a period of not less than one hour. After completion of the test and return to ambient conditions, the unit shall be examined to determine compliance with 3.7.5.4. As an option, individual sub-assembled panels equivalent to a complete unit may be tested in an autoclave to meet the requirements of 3.7.5.4. Panel sets shall be randomly selected from production lots.

4.7.16 Solar load test. The unit, in its shipping configuration, shall be placed in a chamber where the temperature can be stabilized at 100° ± 5°F (37.8 ± 3°C). The test shall begin with both the temperature inside and outside the unit at 75° ± 5°F (23.9 ± 3°C). The temperature in the chamber shall then be raised to 100° ± 5°F (37.8 ± 3°C). An additional solar load sufficient to raise the outer skin temperature to 200° ± 5°F (93.3 ± 3°C) shall be applied to the fixed roof and the folding roof. The solar load shall be applied so that the 200°F (93.3°C) surface temperature is attained in two hours and maintained for an additional four hours. Instrumentation shall be provided so that at least five places can be measured on each surface being tested. At least three places on each surface (including the center of the panels being tested) shall read

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200° ± 5°F (93.3 ± 3°C) during the test. The inside temperature of the unit, measured approximately one foot from the panels being tested, shall be maintained at 75° ± 5°F (23.9 ± 3°C) during the entire period of the test. Upon completion of the solar load test, the outside skin temperature shall be reduced to 75° ± 5°F (23.9 ± 3°C) within two hours. Coin tap testing (see 4.7.4.5) and visual examination shall be performed before and after the test to determine compliance with 3.7.5.5. As an option, the solar load test may be conducted in the expanded position on the fixed and expanding roof panels only. Test temperatures shall be in accordance with those previously stated for the closed configuration test.

4.7.17 Insert torque.

4.7.17.1 Sample panel torque. Torques specified in table I shall be applied to two identical inserts of each size and diameter which have been potted in the sample panel described in 4.6.1.1.4 to determine compliance with 3.7.6.1. Torques shall be applied using a torque wrench and hex head machine screws of a steel alloy which will allow attainment of specified torque without screw failure.

4.7.17.2 Proof torque. A proof torque equal to 80 percent of the applicable torque specified in table I shall be applied to randomly selected inserts of each different size on each panel prior to final assembly to determine compliance with requirements of 3.7.6.1.

4.7.18 Insert pull-out.

4.7.18.1 Sample panel pull-out. Tensile pull-out loads specified in table I shall be applied to two identical inserts of each size and diameter which have been potted in the sample panel described in 4.6.1.1.4 to determine compliance with 3.7.6.1.

4.7.18.2 Proof pull-out. A proof tensile pull-out load equal to 80 percent of the applicable pull-out specified in table I shall be applied to randomly selected inserts (different from those selected for the torque testing of 4.7.17.2) of each different size on each panel prior to final assemblies to determine compliance with requirements of 3.7.6.1.

* 4.7.19 Durability test. The completely fabricated shelter shall be tested for conformance with 3.12. The test shall consist of 6 test cycles of 240 hours each for a total durability test time of 1440 hours and any failure as defined below shall constitute failure of the test. Prior to initiation of the test, the complete shelter shall be unpacked and repacked with a 6500 pound (2.9 Mg) dummy load uniformly distributed on the floor.

a. The test cycle for the multi-purpose shelter shall consist of:

(1) Load the shelter on a transporter dolly set.

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(2) Transport the shelter (in its folded configuration) by transporter dolly set to an erection site, a distance of 25 miles (40.2 km) over cross-country No. 1 (moderate) terrain.
(3) Remove the shelter from the dolly set.
(4) Erect the shelter and test for internal air leakage, 4.7.6.2. The shelter shall remain erect in the operational mode between erection and striking.
(5) Strike the shelter.

b. A failure is defined as any malfunction which the operator cannot remedy by adjustment, repair or replacement action within sixty (60) minutes using tools and parts normally available and which causes or may cause:

(1) Inability to commence operation.
(2) Cessation of operation during the cycle of 240 hours.
(3) Reduction of performance below an acceptable level.
(4) Personnel safety hazard by continued operation.
(5) Serious damage to the shelter or element thereof by continued operation.

Any malfunction attributed to operator error shall not be considered an accountable malfunction.

* 5. PACKAGING

* 5.1 Preservation-packaging. Each shelter with cargo net and straps, struts and complete fabric enclosure placed therein shall have all vents, windows, doors and other openings properly closed, covered and secured by the means provided.

5.2 Packing. Not applicable.

5.3 Marking. In addition to any special marking required by the contract or order, shipments shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. The shelter covered by this specification is intended for use as a container for storing and transporting the Shelter, Air Inflatable, Double Wall, Hospital Ward (MUST) with auxiliary medical equipment and supplies. The unit is also used to house and transport kits of equipment for specific medical support functions such as food service, water treatment and purification units; lavatory, toilet, and shower facilities; for storage of food, clothing and miscellaneous items, and as a shelter for patient receiving, pre-operative and emergency functions.

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6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number and date of this specification.
(b) When a first article is required (see 3.3, 4.3 and 6.3).
(c) When an initial production testing sample is required (see 3.4, 4.4 and 6.4).
(d) When an inspection comparison testing sample is required (see 3.5, 4.5 and 6.5).
(e) When air lock adapter and cargo tie-down assembly are to be Government furnished (see 3.11.1 and 3.11.2).
(f) Source and delivery of Government furnished property (see 3.11.3).

6.3 First article. When a first article is required, it shall be inspected and approved under the appropriate provisions of ASPR 7-104.55. The contracting officer should specify the appropriate type of first article and the number of units to be furnished. The contracting officer should include specific instructions in all procurement instruments regarding arrangements for inspection and approval of the first article.

6.4 Initial production test. When an initial production test sample is required, the contracting officer should include specific instructions in all procurement instruments regarding arrangements for inspection and approval of the initial production test sample.

6.5 Inspection comparison test. When a comparison test sample is required, the contracting officer should include specific instructions on all procurement instruments regarding arrangements for inspection and approval of the comparison test sample.

6.6 Manuals. Any requirements for equipment and instruction manuals for the ward container covered by this specification should be included in DD Form 1423 Contract Data Requirements List and cited in the contract.

6.7 Loading, transporting and railroad humping tests. The contracting officer should make all necessary arrangements with the contractor for testing at a test site sufficiently in advance to minimize the possibility of any delay in testing after the scheduled arrival of the test unit.

6.7.1 Equipment. Equipment required for the transporting test is a standard 2-1/2 or 5-ton cargo truck and a Dolly Set, Lift, Transportable Shelter: Army Model M689E1, (NSN 2330-00-937-1175). Operating supplies, i.e., fuels, oils, etc., shall be furnished by the contractor.

6.8 Changes from previous issue. The margins of this specification are marked with an asterisk (*) to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

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